

POSITIVE LOWER DENSITY IN HOFSTADTER'S $AB - 1$ SEQUENCE

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ABSTRACT. Let $\mathcal{A}$ be the least set of positive integers containing 2 and 3 such that $ab - 1 \in \mathcal{A}$ whenever $a, b \in \mathcal{A}$ are distinct. We prove that $\mathcal{A}$ has positive lower density, answering a question of Hofstadter recorded by Erdős. The proof constructs an injective finite-state family of affine compositions, uses a state-dependent Markov rule to keep the prime exponents of their slopes near a fixed ray, and applies the arithmetic renewal theorem to obtain linearly many distinct values.

1. INTRODUCTION

Let $\mathcal{A}$ be the least subset of $\mathbb{N}$ containing 2, 3 and satisfying

$$a, b \in \mathcal{A}, \quad a \neq b \implies ab - 1 \in \mathcal{A}.$$

Write $A(x) = |\mathcal{A} \cap [1, x]|$. Hofstadter asked whether $\mathcal{A}$ has positive density; Erdős recorded the question in this form in [2], and it is catalogued as Erdős Problem 424 and OEIS A005244 [1, 7]. The later formulation in [3] asks whether almost all integers belong to $\mathcal{A}$, but this is false: induction modulo 3 gives $\mathcal{A} \subseteq \{0, 2\} \pmod{3}$. We prove the positive lower-density statement.

Theorem 1.1. *There is a constant $c > 0$ such that*

$$A(x) \geq cx$$

for all sufficiently large x . Equivalently,

$$\liminf_{x \rightarrow \infty} \frac{A(x)}{x} > 0.$$

The proof uses only the fixed multipliers

$$\mathcal{M} = \{2, 3, 5, 9, 14\}$$

and the affine maps $T_a(x) = ax - 1$. All these multipliers, together with the seed 17, belong to $\mathcal{A}$. We construct many distinct compositions of the T_a having one common slope. A finite family of inverse intervals makes a suitable collection of compositions injective, even though the full affine semigroup is not free; for example, $T_9 \circ T_2 = T_3 \circ T_2 \circ T_3$. Transition probabilities on the interval states are chosen so that a closed path has probability equal to the reciprocal of its slope.

The slopes are products of 2, 3, 5, 7. A state-dependent rule selects among four transition systems so that the three imbalances

$$\nu_2 - 31\nu_7, \quad \nu_3 - 26\nu_7, \quad \nu_5 - 11\nu_7$$

form a recurrent process. First returns to one recurrent state consequently have slopes

$$q^m, \quad q = 2^{31}3^{26}5^{11}7.$$

The first-return paths form an injective code. The arithmetic renewal theorem then gives $\gg q^m$ distinct compositions of slope q^m along an arithmetic progression of exponents. Evaluating them at 17 yields the required elements of $\mathcal{A}$.

Affine closures of recursively generated sets were initiated by Klarner and Rado [5]; see also the survey [6]. A close recent result of Shamazov and Talambutsa proves positive density for affine orbit sets under an exact covering hypothesis [8]. That hypothesis does not hold here, and the semigroup above has relations; the interval construction below instead isolates an injective finite-state subfamily.

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2. A FINITE INTERVAL SYSTEM

We first record why the fixed affine orbit is contained in $\mathcal{A}$.

Lemma 2.1. *If $a_1, \dots, a_r \in \mathcal{M}$, then*

$$T_{a_r} \circ \dots \circ T_{a_1}(17) \in \mathcal{A}.$$

Proof. The identities

$$5 = 2 \cdot 3 - 1, \quad 9 = 2 \cdot 5 - 1, \quad 14 = 3 \cdot 5 - 1, \quad 17 = 2 \cdot 9 - 1$$

show that $\mathcal{M} \cup \{17\} \subseteq \mathcal{A}$, with distinct inputs at every step. If $x \geq 17$ and $a \in \mathcal{M}$, then $a < 17 \leq x$ and $T_a(x) = ax - 1 > x$. Thus a and x are distinct members of $\mathcal{A}$, so $T_a(x) \in \mathcal{A}$. Induction proves the assertion. $\square$

Put $h_a = T_a^{-1}$, so $h_a(x) = (x + 1)/a$. Let

$$K_i = [e_i, e_{i+1}) \quad (1 \leq i \leq 19),$$

where

$$(e_1, \dots, e_{20}) = \left(\frac{1}{9}, \frac{10}{81}, \frac{1}{7}, \frac{1}{5}, \frac{2}{9}, \frac{2}{7}, \frac{1}{3}, \frac{10}{27}, \frac{2}{5}, \frac{11}{27}, \frac{3}{7}, \frac{37}{81}, \frac{5}{9}, \frac{3}{5}, \frac{2}{3}, \frac{19}{27}, \frac{59}{81}, \frac{4}{5}, \frac{23}{27}, 1 \right).$$

For $r \leq s$, write $K_{r:s} = K_r \cup \dots \cup K_s$. The following identities are exact; each row means $T_a(K_i) = K_{r:s}$.

i	a	image	i	a	image
1	14	$K_{13:16}$	11	3	$K_{6:7}$
2	14	$K_{17:19}$	12	3	$K_{8:14}$
2	9	$K_{1:5}$	13	3	$K_{15:17}$
3	9	$K_{6:17}$	13	2	$K_{1:3}$
4	9	$K_{18:19}$	14	3	$K_{18:19}$
5	5	$K_{1:10}$	14	2	$K_{4:6}$
6	5	$K_{11:14}$	15	2	$K_{7:9}$
7	5	$K_{15:18}$	16	2	$K_{10:11}$
8	5	K_{19}	17	2	$K_{12:13}$
8	3	$K_{1:3}$	18	2	$K_{14:15}$
9	3	K_4	19	2	$K_{16:19}$
10	3	K_5			

There are choices at K_2 and K_8 , and at K_{13}, K_{14} we use either multiplier 3 at both states or multiplier 2 at both states. We use the four assignments

$$\alpha_0 = (14, 5, 3), \quad \alpha_1 = (9, 5, 3), \quad \alpha_2 = (14, 3, 3), \quad \alpha_3 = (14, 5, 2),$$

where the entries specify the choices at K_2 , at K_8 , and jointly at K_{13}, K_{14} . Let $a_i^{(\ell)}$ be the multiplier assigned to K_i by α_ℓ . Define a transition matrix $P_\ell = (p_{ij}^{(\ell)})$ by

$$p_{ij}^{(\ell)} = \begin{cases} \frac{|K_j|}{a_i^{(\ell)} |K_i|}, & K_j \subseteq T_{a_i^{(\ell)}}(K_i), \\ 0, & \text{otherwise.} \end{cases}$$

The intervals $h_{a_i^{(\ell)}}(K_j)$ over the allowed successors j partition K_i , so P_ℓ is stochastic. Its directed transition graph is strongly connected; directly from the table, every state reaches 19, and 19 reaches every state.

An allowed path is a sequence

$$\gamma = (i_0, a_0, i_1, a_1, \dots, a_{r-1}, i_r)$$

in which a_t is the multiplier used on leaving K_{i_t} and $K_{i_{t+1}} \subseteq T_{a_t}(K_{i_t})$. Its probability is

$$(1) \quad \mathbb{P}(\gamma) = \prod_{t=0}^{r-1} \frac{|K_{i_{t+1}}|}{a_t |K_{i_t}|} = \frac{|K_{i_r}|}{|K_{i_0}|} \frac{1}{a_0 a_1 \dots a_{r-1}}.$$

The telescoping identity remains valid when the assignment changes along the path.

For a product of multipliers, let

$$\nu = (\nu_2, \nu_3, \nu_5, \nu_7) \in \mathbb{Z}_{\geq 0}^4$$

be its prime-exponent vector, and set

$$H(\nu) = (\nu_2 - 31\nu_7, \nu_3 - 26\nu_7, \nu_5 - 11\nu_7) \in \mathbb{Z}^3.$$

Then $H(\nu) = 0$ exactly when $\nu = m(31, 26, 11, 1)$ for some $m \geq 0$, in which case the product is q^m , where

$$q = 2^{31}3^{26}5^{11}7.$$

For assignment α_ℓ , let $\xi_\ell(i)$ be the increment of H produced by the multiplier $a_i^{(\ell)}$. Let π_ℓ be the stationary distribution of P_ℓ and put

$$d_\ell = \sum_{i=1}^{19} \pi_\ell(i) \xi_\ell(i).$$

Lemma 2.2. *The four stationary drifts are*

ℓ	d_ℓ
0	$\frac{1}{209179}(82502, 60472, 22315)$
1	$\frac{1}{9240032}(4122816, 3176276, 1187745)$
2	$\frac{1}{154297}(-17374, 2952, -12847)$
3	$\frac{1}{938829}(-410774, -384064, -82315).$

The origin lies in the interior of $\text{conv}\{d_0, d_1, d_2, d_3\}$.

Proof. Appendix A records exact stationary distributions, verifies the displayed drifts, and gives a positive affine dependence together with a nonzero determinant certificate. These calculations show that the four drifts are affinely independent and that the origin lies in the interior of their tetrahedron. $\square$

3. A RECURRENT CONTROLLED CHAIN

We now allow the assignment to depend on the accumulated imbalance. The following lemma is the only recurrence input.

Lemma 3.1. *There is a deterministic rule for choosing among $\alpha_0, \dots, \alpha_3$ such that the interval state together with the imbalance vector has a positive recurrent state.*

Proof. Use the Euclidean norm on $\mathbb{R}^3$. For a fixed ℓ , the finite irreducible chain with transition matrix P_ℓ has stationary distribution π_ℓ . Since $\xi_\ell - d_\ell$ has stationary mean zero, the Poisson equation

$$g_\ell - P_\ell g_\ell = \xi_\ell - d_\ell$$

has a solution $g_\ell : \{1, \dots, 19\} \rightarrow \mathbb{R}^3$, coordinatewise. Consequently, for every initial interval state i and every $N \geq 1$,

$$(2) \quad \|\mathbb{E}_i^{(\ell)} \left[\sum_{t=0}^{N-1} \xi_\ell(I_t) \right] - Nd_\ell\| \leq C_0,$$

where $C_0 = 2 \max_{\ell, i} \|g_\ell(i)\|$.

By Lemma 2.2, the origin is an interior point of the convex hull of the drifts. Hence there is $\delta > 0$ such that, for every unit vector $u \in \mathbb{R}^3$,

$$(3) \quad \min_{0 \leq \ell \leq 3} \langle u, d_\ell \rangle \leq -\delta.$$

Choose a block length N , to be fixed below. At time kN , if the current imbalance is $h \neq 0$, choose an index $\ell(h)$ minimizing $\langle h/\|h\|, d_\ell \rangle$ and use assignment $\alpha_{\ell(h)}$ for the next N steps. Use α_0 when $h = 0$, and fix a deterministic rule for ties. Then

$$X_k = (I_{kN}, H_{kN})$$

is a time-homogeneous Markov chain on $\{1, \dots, 19\} \times \mathbb{Z}^3$.

Put $M_0 = \max_{\ell, i} \|\xi_\ell(i)\|$. If the block increment is $Z = H_{(k+1)N} - H_{kN}$, then $\|Z\| \leq M_0 N$. For $h \neq 0$ we use the elementary inequality

$$\|h + z\| - \|h\| \leq \left\langle \frac{h}{\|h\|}, z \right\rangle + \frac{\|z\|^2}{2\|h\|},$$

After adding $\|h\|$, the right-hand side is nonnegative; squaring reduces the claim to a square. Combining this with (2) and (3) gives

$$(4) \quad \mathbb{E}[\|H_{(k+1)N}\| - \|H_{kN}\| \mid X_k = (i, h)] \leq -N\delta + C_0 + \frac{M_0^2 N^2}{2\|h\|}.$$

Choose N so that $N\delta > C_0 + 2$, and then choose R so large that the right-hand side of (4) is at most -1 whenever $\|h\| > R$. Let

$$C = \{(i, h) : 1 \leq i \leq 19, h \in \mathbb{Z}^3, \|h\| \leq R\}.$$

This set is finite. If $\tau_C = \inf\{k \geq 0 : X_k \in C\}$, the drift estimate and optional stopping applied to $\|H_{kN}\|$ give

$$(5) \quad \mathbb{E}_{(i, h)} \tau_C \leq \|h\| \quad ((i, h) \notin C).$$

Indeed, apply the drift inequality to $\tau_C \wedge n$ and let $n \rightarrow \infty$.

For $x = (i, h) \in C$, put $\tau_C^+ = \inf\{k \geq 1 : X_k \in C\}$. After one block the imbalance has norm at most $R + M_0 N$, so by the Markov property and (5),

$$\mathbb{E}_x \tau_C^+ \leq 1 + \mathbb{E}_x [\mathbf{1}_{\{X_1 \notin C\}} \mathbb{E}_{X_1} \tau_C] \leq 1 + R + M_0 N.$$

Thus successive visits to C define a Markov chain on a finite state space, with uniformly bounded expected excursion lengths in the original block chain. Choose a state s in a recurrent class of this embedded finite chain. Its expected return time in the embedded chain is finite; summing the uniformly bounded expected excursion lengths shows that its return time in (X_k) also has finite mean. Hence s is positive recurrent. $\square$

Fix a positive recurrent state

$$s = (i_*, h_*),$$

and start the controlled chain from s .

4. RETURN PATHS AND RENEWAL

Let

$$\tau = \inf\{k \geq 1 : X_k = s\}$$

be the first return time in block units. A realization of this return determines an underlying path of length $L = N\tau$,

$$\gamma = (i_0, a_0, i_1, a_1, \dots, a_{L-1}, i_L), \quad i_0 = i_L = i_*.$$

Let $\mathcal{R}$ be the set of all such first-return paths. Define

$$G_\gamma = T_{a_{L-1}} \circ \dots \circ T_{a_0}, \quad Q_\gamma = a_0 a_1 \dots a_{L-1},$$

and

$$J_\gamma = G_\gamma^{-1}(K_{i_*}) \subseteq K_{i_*}.$$

Lemma 4.1. *Distinct finite concatenations of paths from $\mathcal{R}$ give distinct affine maps.*

Proof. No first-return path is a proper initial segment of another. Consider two distinct paths in $\mathcal{R}$, and let their first differing transition occur after a common path prefix. Starting from the same full state s , the common prefix gives the same accumulated imbalance, so the deterministic rule chooses the same multiplier at the point of divergence. The two paths enter distinct successor intervals. Their corresponding inverse branches $h_a(K_j)$ are disjoint subsets of the current interval, and pulling them back through the common preceding composition shows that the intervals J_γ for the two paths are disjoint. Equality of their affine maps would give equal inverse images of K_{i_*} , which is impossible.

Now consider two concatenations. Delete their common initial return paths. If both have a return path remaining, their full inverse images of K_{i_*} lie, after pulling back through the common prefix, inside the disjoint intervals associated with their first different return paths. The maps cannot be equal. If only one concatenation has a path remaining, its slope is strictly larger than that of the other, since every multiplier exceeds one. Thus the maps are distinct in all cases. $\square$

For $\gamma \in \mathcal{R}$, equation (1) and $i_0 = i_L$ give

$$(6) \quad \mathbb{P}(\gamma) = Q_\gamma^{-1}.$$

The return also restores the imbalance coordinate. Hence the net prime-exponent vector of Q_γ lies in the kernel of H , so

$$Q_\gamma = q^m$$

for a unique integer $m \geq 1$.

Let b_m be the number of first-return paths of slope q^m . Positive recurrence implies $\tau < \infty$ almost surely, and the first-return events are disjoint, so (6) gives

$$(7) \quad \sum_{m \geq 1} b_m q^{-m} = 1.$$

If such a path has length L , each multiplier contributes one or two prime factors counted with multiplicity, whereas q^m has $69m$ prime factors. Therefore

$$(8) \quad L \leq 69m \leq 2L.$$

In particular, each b_m is finite, since the interval graph has only finitely many paths of bounded length. If M is the exponent of the random first-return path, then $L = N\tau$ and

$$(9) \quad \mathbb{E}M = \sum_{m \geq 1} m b_m q^{-m} \leq \frac{2N}{69} \mathbb{E}\tau < \infty.$$

Proposition 4.2. *There are an integer $d \geq 1$ and a constant $\eta > 0$ such that, for all sufficiently large r , at least ηq^{rd} distinct concatenations of first-return paths have slope q^{rd} .*

Proof. Put $p_m = b_m q^{-m}$. By (7) and (9), $(p_m)_{m \geq 1}$ is a probability distribution of finite mean

$$\mu = \sum_{m \geq 1} m p_m.$$

Successive returns to s have independent, identically distributed exponents by the strong Markov property. Let c_n be the number of ordered concatenations of return paths whose total exponent is n , including the empty concatenation at $n = 0$. Every such concatenation has probability q^{-n} , and hence

$$(10) \quad c_n q^{-n} = \sum_{k \geq 0} p^{*k}(n).$$

Let $d = \gcd\{m : p_m > 0\}$. The arithmetic renewal theorem gives

$$\sum_{k \geq 0} p^{*k}(rd) \longrightarrow \frac{d}{\mu} \quad (r \rightarrow \infty)$$

[4, Chapter XI, Section 1]. Taking $\eta = d/(2\mu)$, equation (10) gives $c_{rd} \geq \eta q^{rd}$ for all sufficiently large r . Lemma 4.1 makes the corresponding affine maps distinct. $\square$

Proof of Theorem 1.1. Every nonempty composition of the maps T_a has the form

$$G(x) = Qx - C$$

with integers Q, C satisfying $0 < C < Q$. This follows by induction: it is true for T_a , and

$$T_a(Qx - C) = aQx - (aC + 1), \quad 0 < aC + 1 < aQ.$$

Fix a sufficiently large r and put $Q = q^{rd}$. Proposition 4.2 gives at least ηQ distinct compositions of slope Q . They have distinct constant terms by Lemma 4.1, and therefore take distinct values at 17. Lemma 2.1 puts all these values in $\mathcal{A}$, while $0 < C < Q$ gives

$$16Q < G(17) = 17Q - C < 17Q.$$

Consequently,

$$A(17q^{rd}) \geq \eta q^{rd}$$

for all sufficiently large r . Given sufficiently large x , choose r with

$$17q^{rd} \leq x < 17q^{(r+1)d}.$$

Then

$$\frac{A(x)}{x} \geq \frac{\eta q^{rd}}{17q^{(r+1)d}} = \frac{\eta}{17q^d} > 0.$$

This proves the theorem. □

APPENDIX A. EXACT STATIONARY DATA

For $0 \leq \ell \leq 3$, write $\pi_\ell = w^{(\ell)}/D_\ell$. The denominators are

$$(D_0, D_1, D_2, D_3) = (5647833, 83160288, 6943365, 76045149),$$

and the integer weights are listed below.

i	$w_i^{(0)}$	$w_i^{(1)}$	$w_i^{(2)}$	$w_i^{(3)}$
1	7350	126000	48510	877590
2	11550	198000	76230	1379070
3	34020	583200	224532	4061988
4	66150	1008045	87318	2296350
5	189000	2880000	239400	6561000
6	145125	2160000	201825	5323725
7	264600	3910725	352800	3483900
8	211680	3124980	282240	2787120
9	52920	781245	70560	696780
10	151200	2232000	191520	1990800
11	202000	2976000	258000	2702160
12	707000	10380000	903000	9457560
13	320040	4703400	418824	4481568
14	490644	7211349	628236	6841800
15	303450	4461450	391650	2487450
16	202300	2974000	237580	1658300
17	586670	8572400	688982	4809070
18	441294	6450654	498918	3668238
19	1260840	18426840	1143240	10480680

Each column sums to the corresponding D_ℓ . Substitution in the transition matrices defined in Section 2 gives

$$w^{(\ell)} P_\ell = w^{(\ell)}.$$

Thus the normalized columns are the stationary distributions. The multiplier increments are

$$\xi(2) = (1, 0, 0), \quad \xi(3) = (0, 1, 0), \quad \xi(5) = (0, 0, 1),$$

$$\xi(9) = (0, 2, 0), \quad \xi(14) = (-30, -26, -11).$$

Summing these increments against the four stationary distributions gives the drift vectors in Lemma 2.2.

For the convex-hull assertion, let

$$(c_0, c_1, c_2, c_3) = (90039171528246091, 305527530231216384, \\ 237978342976556325, 331488754119440400).$$

All four coefficients are positive, and exact arithmetic gives

$$c_0d_0 + c_1d_1 + c_2d_2 + c_3d_3 = 0,$$

while

$$\det(d_1 - d_0, d_2 - d_0, d_3 - d_0) = \frac{203470981686392000}{105416226360733049619} \neq 0.$$

Thus the origin is an interior point of the tetrahedron spanned by the four drifts. All verifications in this appendix use exact integer and rational arithmetic.

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